

A Hands-On Introduction to Machine Learning



Julian Gold
Jonathan Hanke

January 13–16, 2026



With materials from:

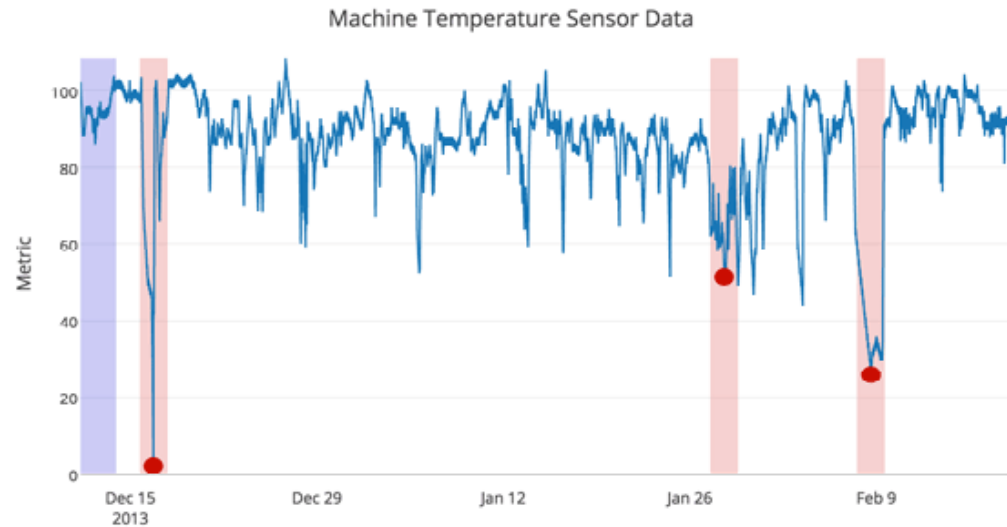
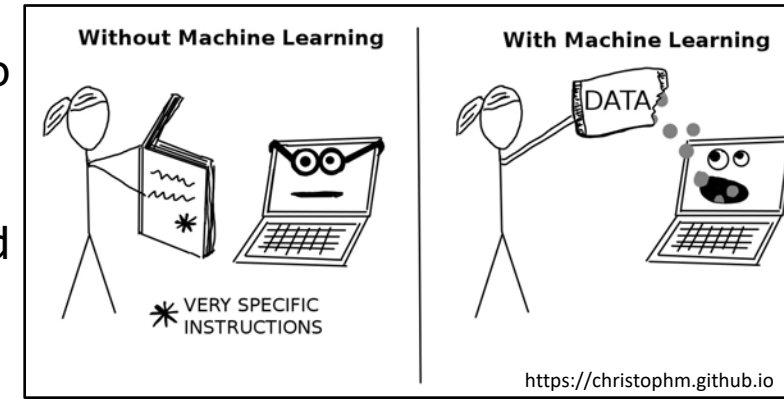
Brian Arnold, Gage DeZoort, Julian Gold, Jonathan Halverson, Jonathan Hanke, Christina Peters
Jake Snell, Savannah Thias, Amy Winecoff

Mini-Course Outline

Date	Topic	Instructor
Tue. 1/13	Machine Learning Overview and Simple Models	Julian Gold
Wed. 1/14	Model Evaluation and Improving Performance	Julian Gold
Thu. 1/15	Introduction to Neural Networks	Jonathan Hanke
Fri. 1/16	Survey of Neural Network Architectures	Jonathan Hanke

What is machine learning?

1. building and understanding methods that 'learn' by using data to improve performance on some set of tasks
2. using and developing computer systems that can learn and adapt without following explicit instructions, by using algorithms and statistical models to analyze and draw inferences from patterns in data.



Also known as:

- pattern recognition
- artificial intelligence
- data mining
- predictive analytics

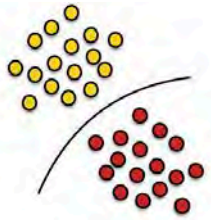
Goal is often to use data to create an algorithm/model that

- makes accurate predictions
- is interpretable, revealing (previously unknown) patterns in data

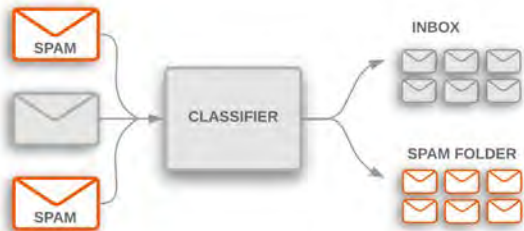
ML tasks

1. building and understanding methods that 'learn' by using data to improve performance on some set of tasks

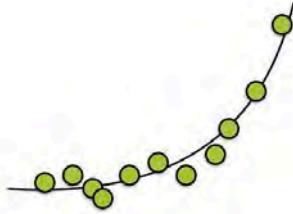
classification



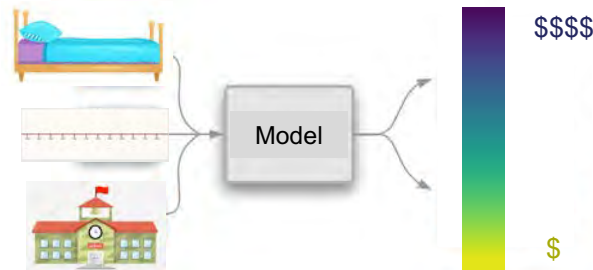
predict input data
labels



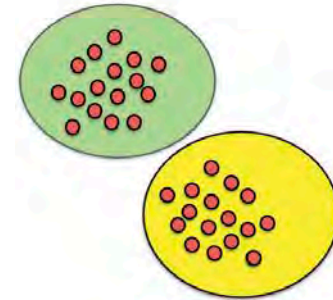
regression



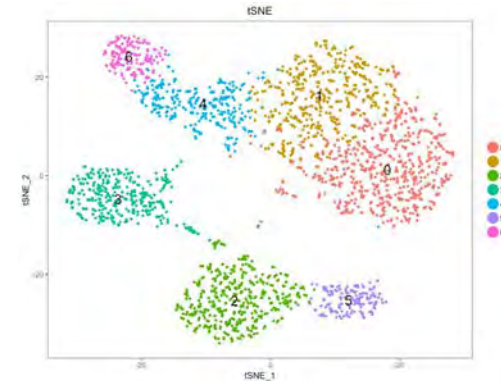
predict input data
values



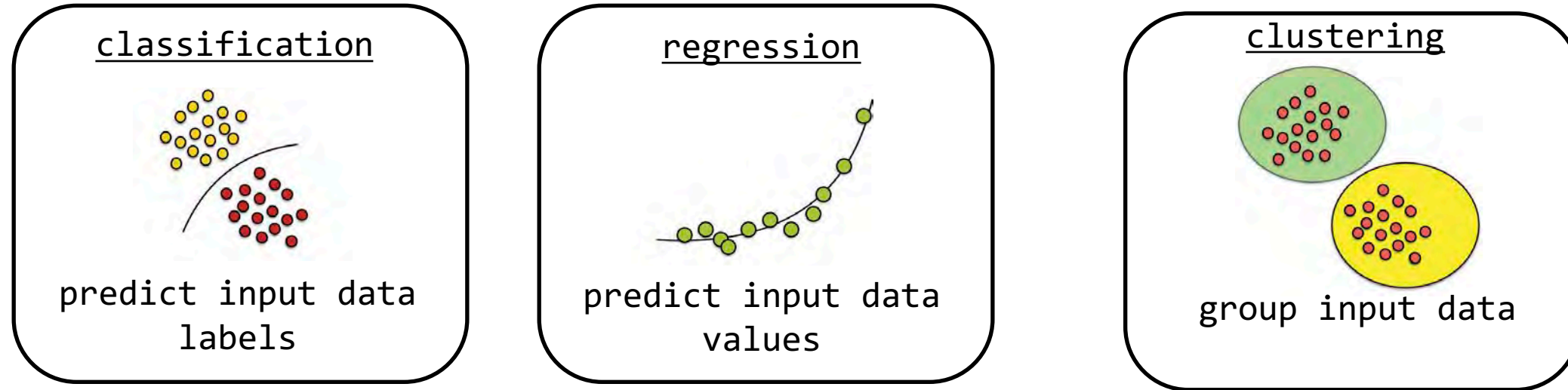
clustering



group input data

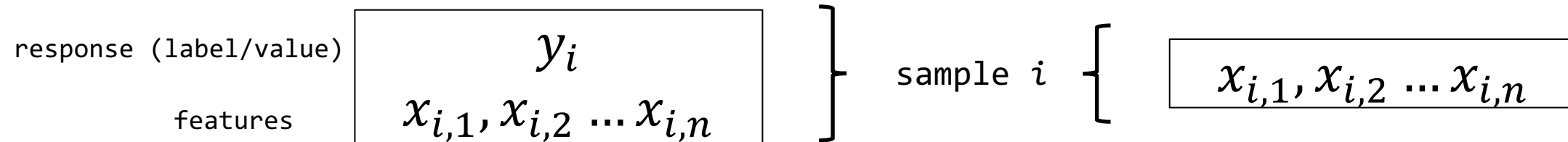


ML tasks



supervised

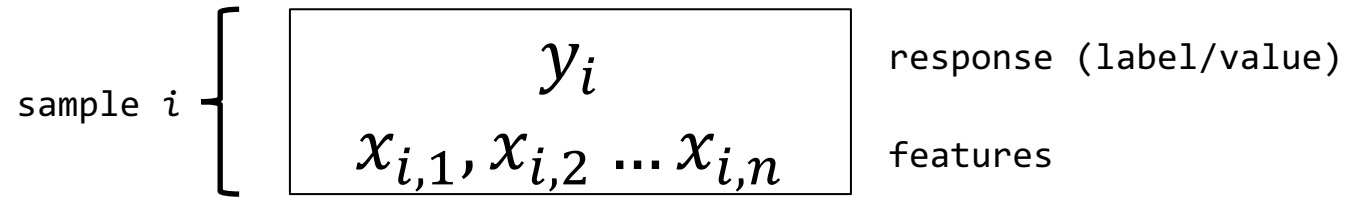
unsupervised



ML learns the relationship
between the features and
response

ML learns patterns/groupings

Terminology



sample i

- sample
- data point
- observation

y_i

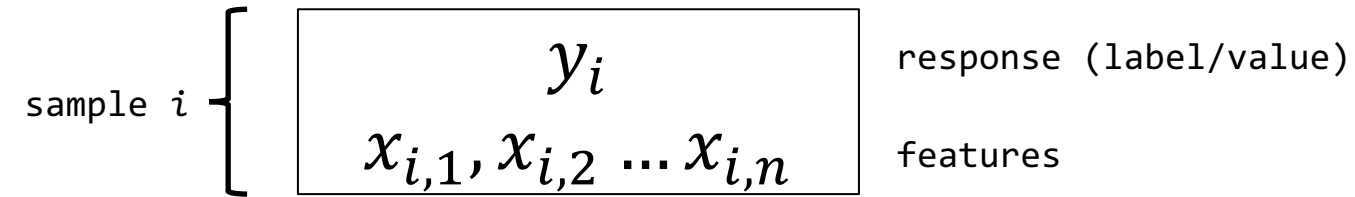
- response
- target
- class (if categorical)
- outcome
- dependent variable

$x_{i,1}, x_{i,2} \dots x_{i,n}$

- features
- predictors
- descriptors
- attributes
- covariates
- independent variables

many terms in English, but the math is always the same!

Discrete/categorical or continuous values!



examples

response y_i

- a sample's disease status (discrete)
- a sample's height/length (continuous)
- a house's market value (continuous)

features $(x_{i,1}, x_{i,2} \dots x_{i,n})$

- the presence of a mutation in genome (discrete)
 - cigarettes smoked per week (continuous)
 - the age of a house (continuous)
-

Why use machine learning?

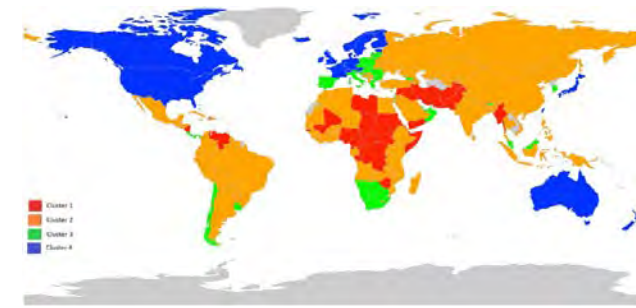
we want to

- know a future event
- make decision based on information
- look for useful patterns in data



examples

- supervised
 - should I sell this stock?
 - how many copies will this book sell?
 - will this customer move their business to a different company?
 - how much will my house sell for in the current market?
 - does a patient have a specific disease?
 - based on past choices, which movies will interest this viewer?
 - which people should we match in our online dating service?
 - will this patient respond to this therapy?
- unsupervised
 - how do customers differ from one another?
 - how are countries different in terms of socio-economic/health?
 - how many cell types are in my sample?



Why use machine learning?

supervised

- prediction
 - predict response value for new samples
- inference
 - understand *how* and *why* a model works

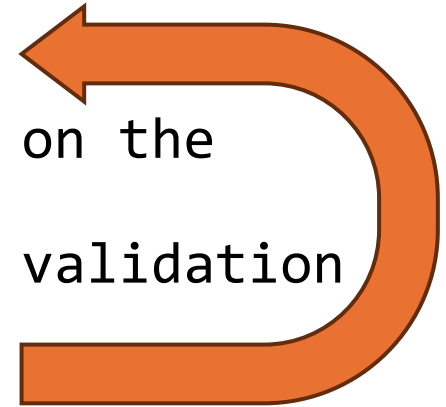
unsupervised

- learn underlying structure of data
-

Overview of Machine Learning Process

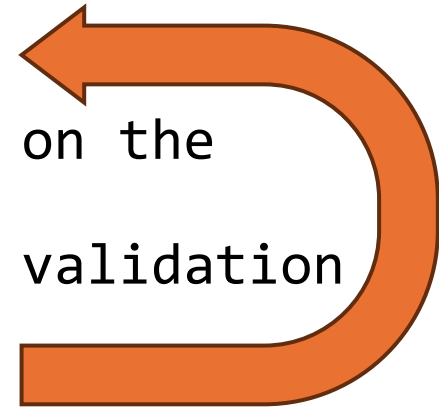
1. Define the problem to be solved.
2. Split the data into train / validation / test.
3. Run the validation loop:
 - a. Choose a set of models.
 - b. Train each model by optimizing its parameters on the training set.
 - c. Evaluate the performance of each model on the validation set.
 - d. Repeat until performance is satisfactory.
4. Evaluate final performance on the test set.

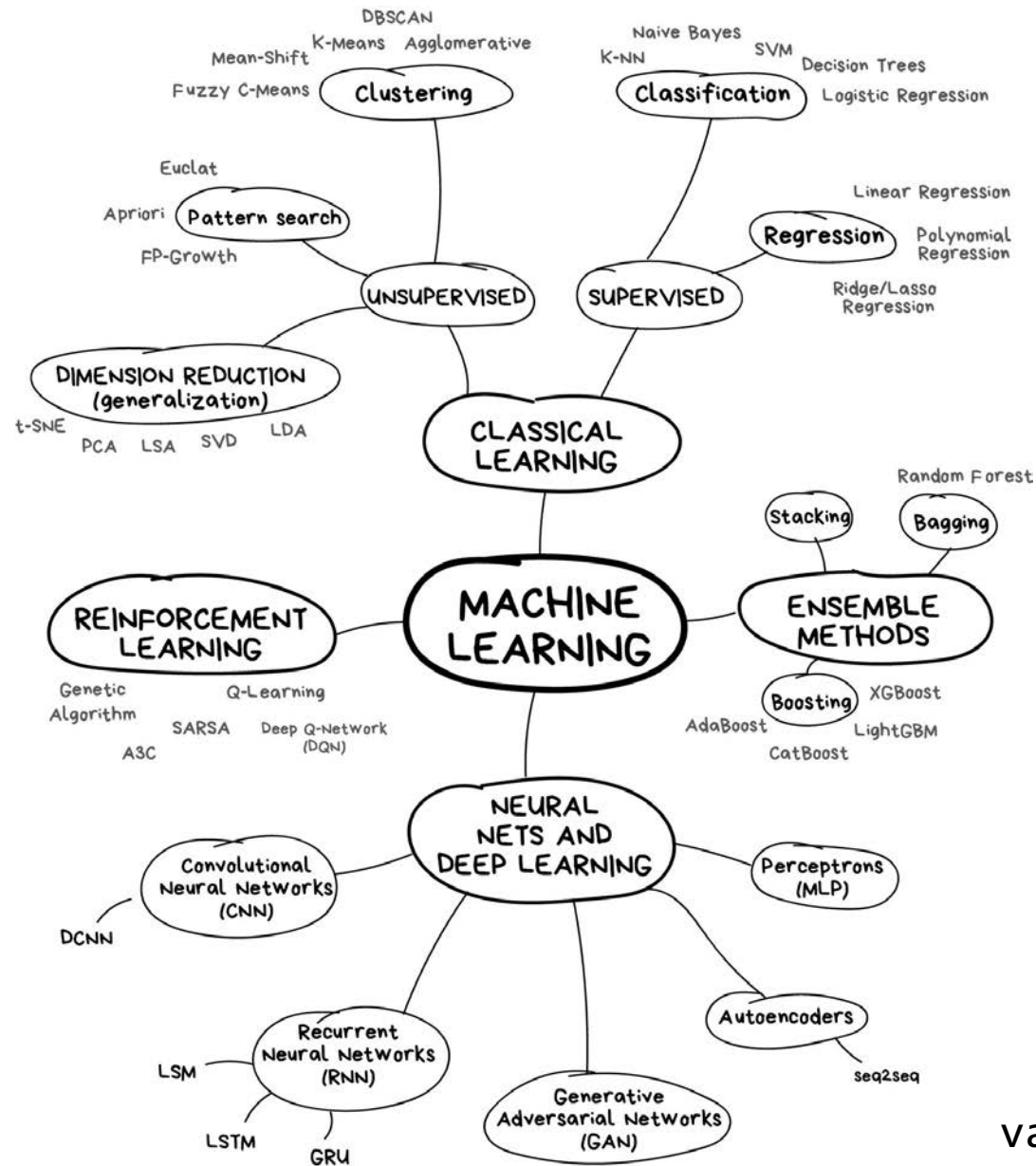
Datasets? Input features?
Targets? Evaluation metrics?



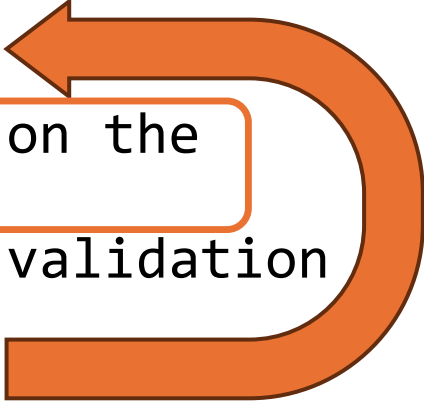
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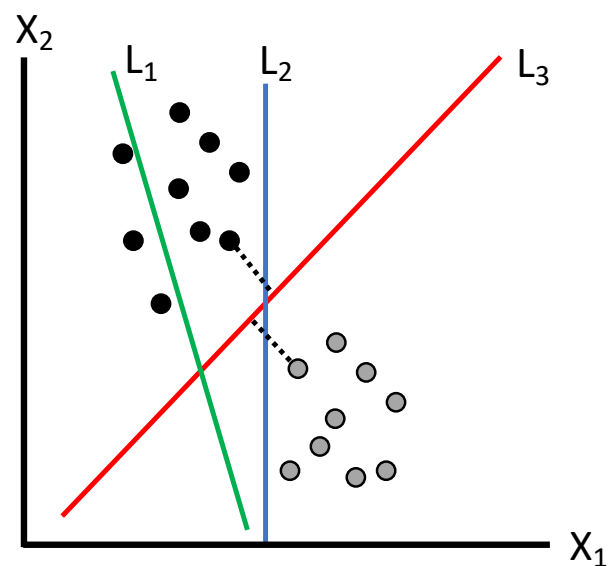


Overview of Machine Learning Process

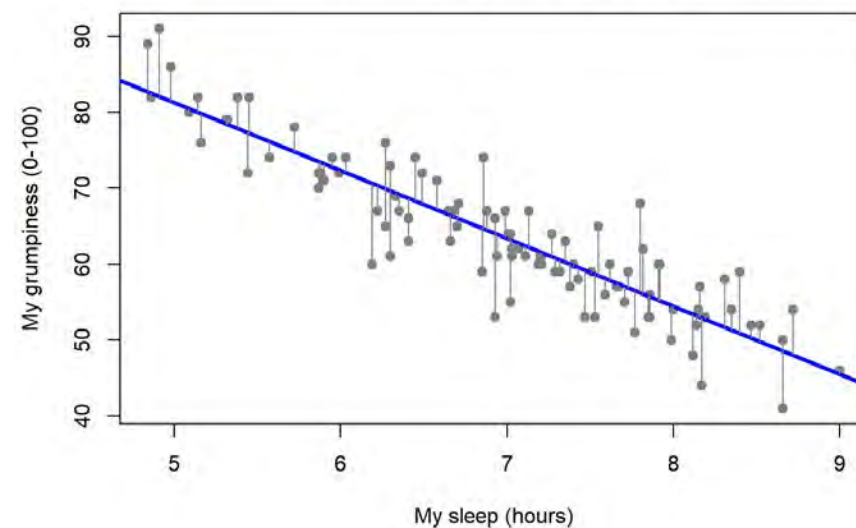
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Model Training

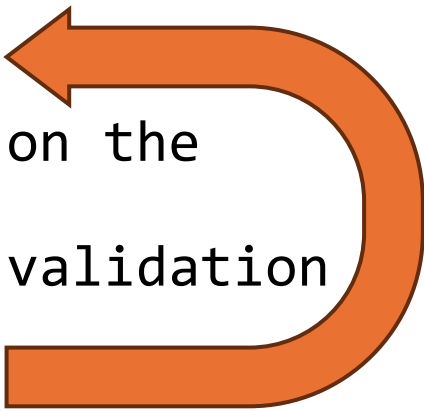
e.g. find slope of line that **best** separates training labels



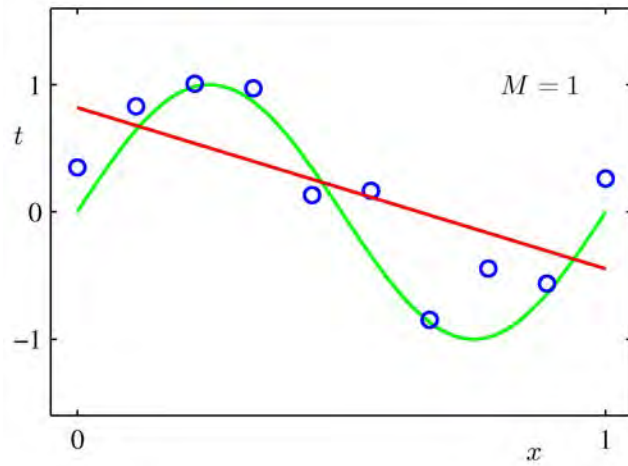
e.g. find slope of line that **best** predicts training values



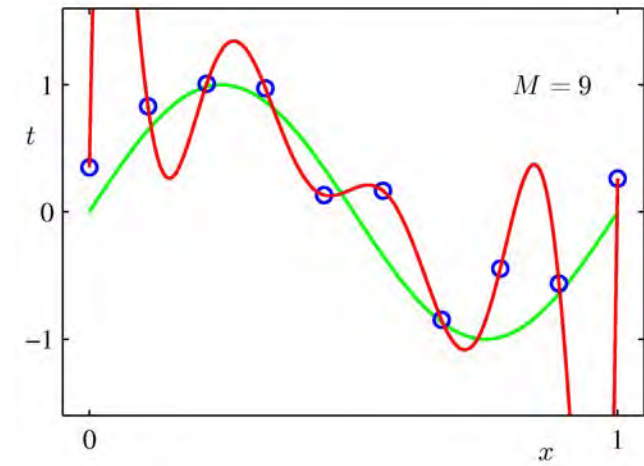
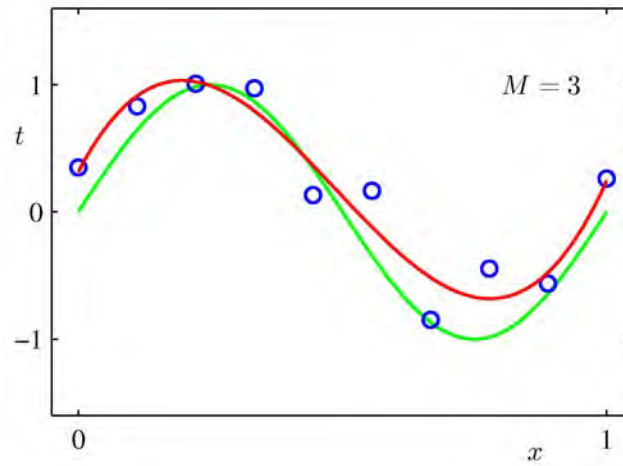
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Model Complexity



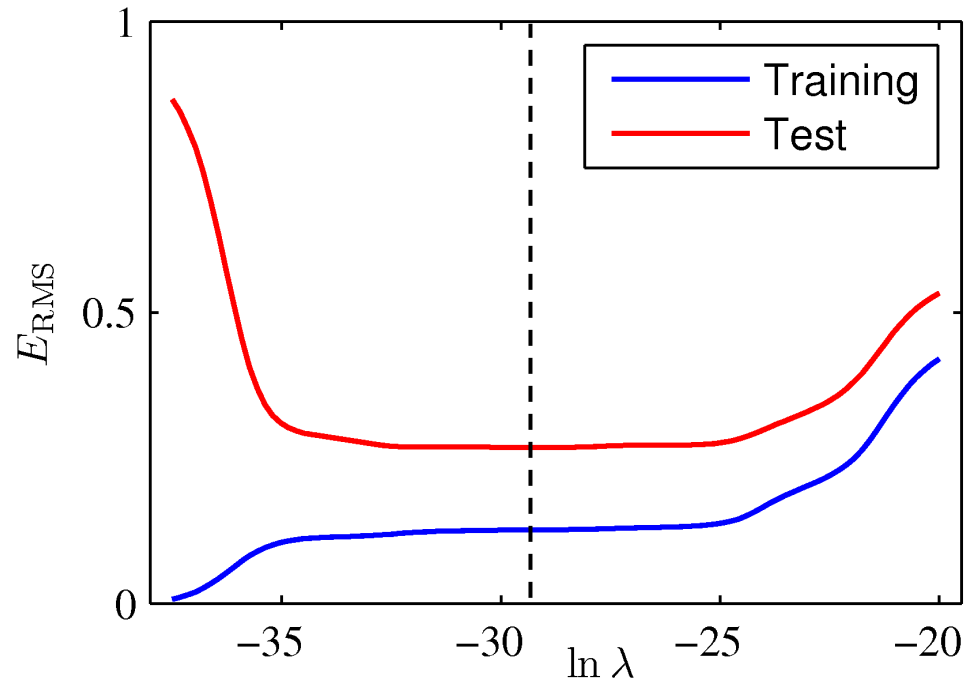
Underfitting
(misses general trend)



Overfitting
(captures noise)



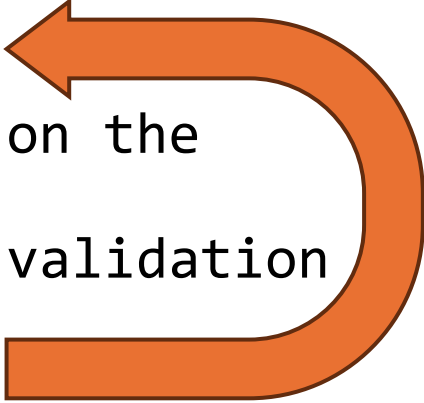
Effect of Complexity on Test Performance



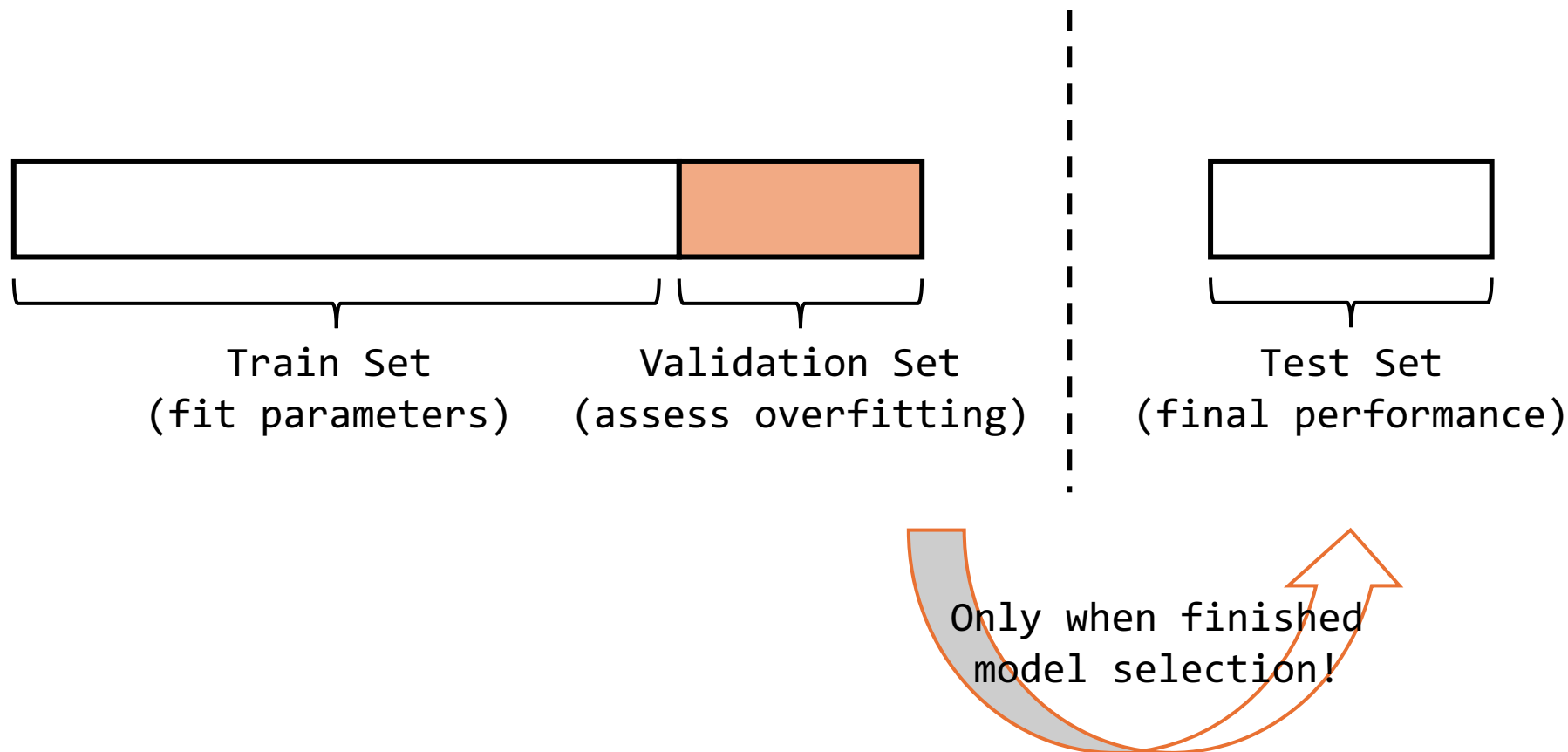
Best model has this value!

Figure credit: Bishop, Christopher M. 2006. *Pattern Recognition and Machine Learning*.

Overview of Machine Learning Process

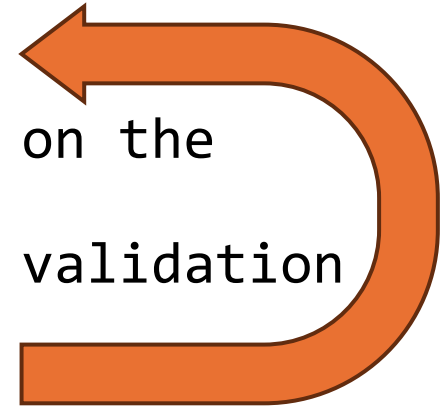
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Role of the Validation Set

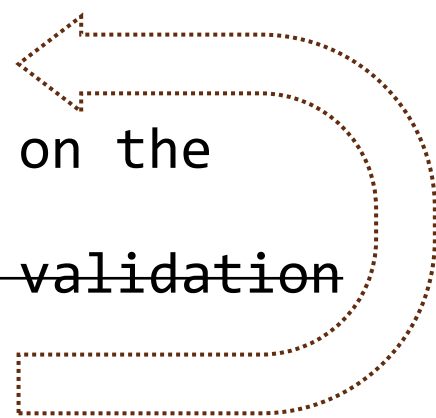


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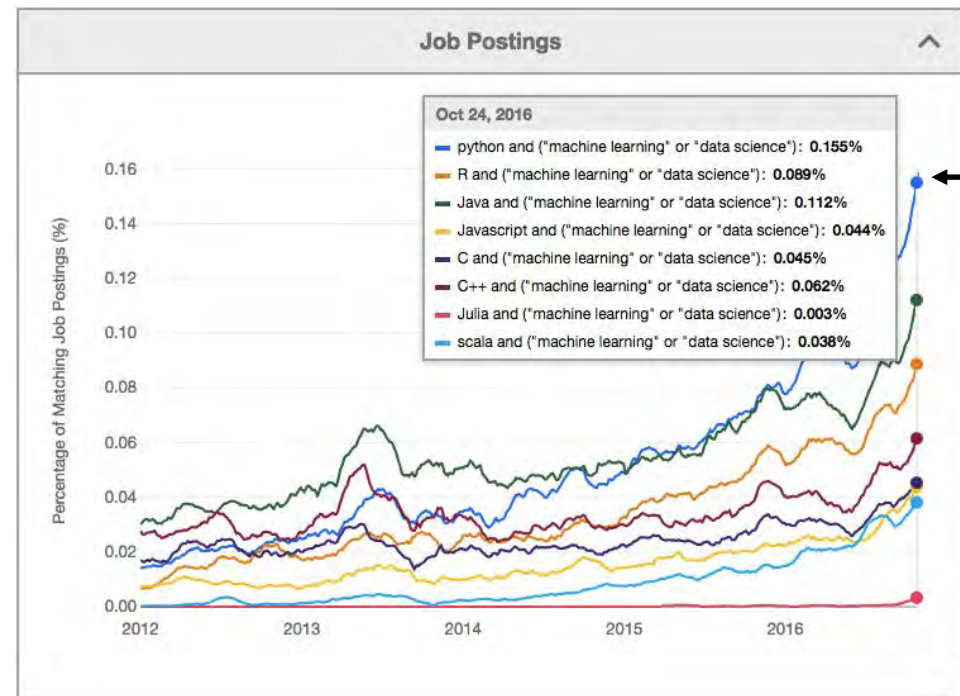
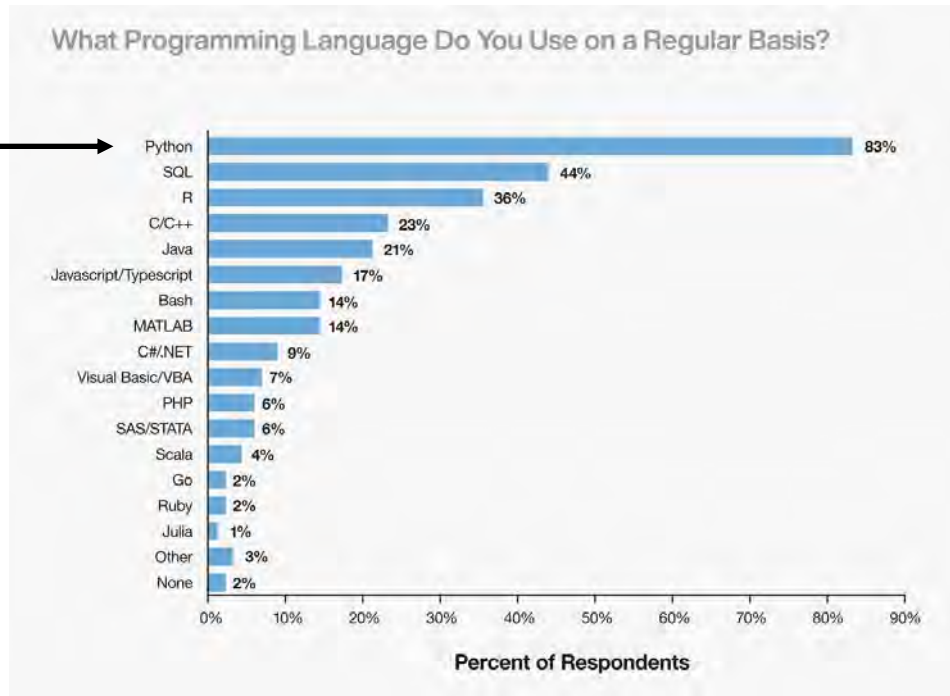
Simplified Machine Learning Process

1. Download a dataset from the internet.
 2. Use the predefined train / test split.
 - ~~3. Run the validation loop:~~
 - ~~a. Choose a set of models.~~
 - ~~b. Train each model by optimizing its parameters on the training set.~~
 - ~~c. Evaluate the performance of each model on the validation set.~~
 - ~~d. Repeat until performance is satisfactory.~~
 4. Evaluate final performance on the test set.
- 

Machine learning in pythonTM

- several options for building ML models
- Python most popular and most in demand (job postings)
- R also popular in statistics and biology communities

python →



← python

Fantastic Python libraries

- data analysis
 - Pandas: great for analyzing and manipulating data tables
 - Seaborn: simple functions -> detailed visualization, integrated with Pandas
 - Matplotlib: visualization



- machine learning
 - numpy: fast, powerful data structures for matrices
 - scikit-learn: simple, efficient, accessible tools for ML
 - Keras: neural networks
 - TensorFlow
 - PyTorch
 - ...

today we will use **numpy** and **scikit-learn**!

ML Coding Tour in Python!



Open the iPython
notebook from this link!

https://github.com/PrincetonUniversity/intro_machine_learning/tree/main/day1

<https://qrco.de/bgYIM1>

Intro to K -nearest neighbors (KNN)

- simple but powerful
- can be used for classification *or* regression!
- algorithm
 1. for a given test sample (yellow dot), find the K nearest training samples in feature space
 - 2a. for **classification**, assign label by majority vote
 - 2b. for **regression**, assign value by mean of neighbors

K is a tunable parameter!

- choose value that gives better predictions on test data

